Probability, Lecture 7-8, reminder

Theorem. Law of Large Numbers (LLN). Let $X_1, X_2, X_3, ...$ be a sequence of independent identically distributed random variables with finite means μ. Their partial sums satisfy

$$\frac{1}{n}(X_1 + X_2 + ... + X_n) \to \mu$$

in distribution, as $n \to \infty$.

(That is, the distribution of the partial sums converge to a distribution concentrated on the single point μ. This means, the distribution functions of the partial sums converge to a step function, having one jump, at μ.)

Theorem. Central Limit Theorem (CLT). Let $X_1, X_2, X_3, ...$ be a sequence of independent identically distributed random variables with finite means μ and finite non-zero variances σ^2, and let $S_n = X_1 + X_2 + ... + X_n$. Then

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \to N(0, 1)$$

in distribution, as $n \to \infty$.

This means, if we divide the sum by $\sqrt{n}$ or $\sqrt{n\sigma^2}$ instead of n (which was done in LLN), it won't be concentrated to one point any more: after subtracting the mean, the limiting distribution will be Normal (Gaussian). There are two interpretations to the formula $\frac{S_n - n\mu}{\sqrt{n\sigma^2}}$ and the theorem:

1) We consider the *sum* of $X_1, X_2, X_3, ...$ ($= S_n$), subtract its mean ($n\mu$), then divide by its standard deviation ($\sqrt{n}\sigma$). The limit when $n \to \infty$ will be Standard Normal in distribution.

2) We consider the *average* of $X_1, X_2, X_3, ...$ ($= \frac{S_n}{n}$), subtract *its* mean (μ), then divide by *its* standard deviation ($\frac{\sigma}{\sqrt{n}}$). The limit when $n \to \infty$ will be Standard Normal in distribution.

To understand this, we need to consider the *standard deviation of the sum of independent random variables.* Using that $E(XY) = E(X)E(Y)$ holds for *independent* random variables, the variances will simply add up in this case (I leave it as a homework). So

$$Var(X_1 + X_2 + ... + X_n) = Var(X_1) + Var(X_2) + ...Var(X_n)$$

if $X_1, X_2, ..., X_n$ are independent. Therefore the variance of the sum of an i.i.d. (independent identically distributed) sequence of random variables is n times the individual variance. We also know that $Var(cX) = c^2Var(X)$ for any constant c. So

$$Var\left(\frac{X_1 + X_2 + ... + X_n}{n}\right) = \frac{Var(X_1 + X_2 + ... + X_n)}{n^2} = \frac{nVar(X_1)}{n^2} = \frac{Var(X_1)}{n}.$$

So the standard deviation of the average is $\frac{SD(X_1)}{\sqrt{n}}$.

Example (using the variance of an independent sum): The capacity of an elevator is 320 kg. People's weight can be regarded as Normally distributed random variables with mean 75 and variance 12^2. What is the probability that four people together, with independent weights, will not overload the lift?

For the solution we need to know that the sum of independent Normally distributed random variables is Normally distributed as well. Its mean will be the sum of the original means, and its variance will be the sum of the original variances. So, if we use the notation X_i for the weight of the ith person, $(X_1 + X_2 + X_3 + X_4) \sim N(4 \times 75,\ 4 \times 12^2) = N(300,\ 24^2)$.

Now we apply a useful transformation (*standardisation*) that we also used in the Central Limit Theorem. The probability that the group of four people will be overweight is

$$P(X_1 + X_2 + X_3 + X_4 > 320) = P(X_1 + X_2 + X_3 + X_4 - 300 > 20) = P(\frac{X_1 + X_2 + X_3 + X_4 - 300}{24} > \frac{20}{24}).$$

We subtracted the mean of the sum, so the new mean will be zero. Also, we divided the expression on the left hand side by the standard deviation, so the new standard deviation will be 1. The distribution remains

Normal, so $\frac{X_1+X_2+X_3+X_4-300}{24}$ will have the Standard Normal distribution. The expression we gained, is almost the cumulative distribution function of this expression at $\frac{20}{24}$, but with reversed inequality sign. So

$$P(\frac{X_1+X_2+X_3+X_4-300}{24} > \frac{20}{24}) = 1 - P(\frac{X_1+X_2+X_3+X_4-300}{24} \leq \frac{20}{24}) = 1 - \Phi(\frac{20}{24})$$

where Φ is the Standard Normal cumulative distribution function, its values are tabulated, see for example http://www.thestudentroom.co.uk/attachment.php?attachmentid=134337&d=1330530156.

From this table, $\Phi(\frac{20}{24}) = \Phi(0.83) = 0.7967$, so $P(X_1 + ... + X_4 > 320) = 1 - 0.7967 = 0.2033 = 20\%$. The probability that four people won't fit into the lift is about 20 percent.

We can not only add variables of the same variance, but also variables of different variances. *Example:* if $X \sim N(-1, 2^2)$ and $Y \sim N(5, 3^2)$ then the mean of $Z = 3X - 4Y$ is $3 \times (-1) - 4 \times 5 = -3 - 20 = -23$ (summing the means) and its variance is $Var(3X) + Var(-4Y) = 3^2 Var(X) + (-4)^2 Var(Y) = 9 \times 2^2 + 16 \times 3^2 = 36 + 144 = 180$. (Note that the variances will add up, regardless of summing or subtracting the variables!) So $Z \sim N(-23, 180)$.

LLN and CLT both can be proven elegantly using characteristic functions (you can look up the proofs in the Grimmett-Stirzaker book). I only mention here that the characteristic function of the Normal distribution is $E(e^{itX}) = e^{it\mu - \frac{1}{2}t^2\sigma^2}$. Specifically $E(e^{itX}) = e^{-t^2/2}$ for the Standard Normal distribution.

(For the Standard Normal distribution, $\phi(t) = E(e^{itX}) = \int_{-\infty}^{\infty} e^{itx} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$. Instead of this complex integral, we can integrate the following: $M(s) = E(e^{sX}) = \int_{-\infty}^{\infty} e^{sx} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{sx - \frac{1}{2}x^2} dx = e^{s^2/2}$. (This is Homework 9.) Then, we may not substitute $s = it$ without justification. In this particular instance the theory of analytic continuation of functions of a complex variable provides this justification (more advanced maths...), and we deduce that $\phi(t) = E(e^{itX}) = e^{-t^2/2}$.)

(If X and Y are independent, the characteristic function of $X + Y$ is the product of the individual characteristic functions: $\phi_{(X+Y}(t) = E(e^{it(X+Y)}) = E(e^{itX})E(e^{itY}) = \phi_X(t)\phi_Y(t)$. Using the above given form of the Normal characteristic function, easily follows that the sum of two independent Normally distributed random variables is Normally distributed, too.)

Estimation, confidence intervals and hypothesis tests

The Central Limit Theorem states that, for large (i.i.d.) samples, $\bar{X}$ (the average) has the distribution $N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately, where $\mu = E(X_1)$, $\sigma^2 = Var(X_1)$, and n is the sample size. This will enable us to give confidence intervals around our estimations, and also testing hypotheses based on these intervals.

The 95% *confidence interval* for the mean μ is

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}.$$

Single-sample problems:

(i) If a random (i.i.d.) sample $X_1, X_2, ..., X_n$ is selected from $N(\mu, \sigma^2)$ then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

for any $n \in \mathbb{N}$, and

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1^2).$$

For large n we can use $S = \sqrt{\frac{1}{n-1}\sum(X - \bar{X})^2} = \sqrt{\frac{1}{n-1}(\sum X^2 - n\bar{X}^2)}$ as the estimation of σ, and approximately

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1^2).$$

(ii) If $X_1, X_2, ..., X_n$ is selected from a population (not necessarily Normal) with mean μ and variance σ^2 then for large n approximately

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1^2)$$

by the Central Limit Theorem, and also

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim N(0,1^2).$$

(iii) *Two-sample problems:* if random samples $X_1, X_2, ..., X_{n_1}$ and $Y_1, Y_2, ..., Y_{n_2}$ are selected, with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively, then under suitable conditions, approximately

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}} \sim N(0,1^2),$$

and for large samples

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}} \sim N(0,1^2).$$

Hypothesis-testing based on the above propositions (using CLT)

The principle is that we calculate the numerical expression specified above in each case, called *statistic*, and check whether it is acceptable according to the *distribution* the statistic should follow, as stated above.

For example, suppose we want to know the mean hight of a population (classroom, village, girls, etc.). Someone told us (and this will be our hypothesis, H_0) that the mean is 172 cm. We take a sample, and calculate the expression mentioned above in cases (i) and (ii): the sample mean minus 172, divided by the standard deviation of the mean, that is either $\sigma/\sqrt{n}$ if σ is given, or estimated from the sample ($S/\sqrt{n}$, see the definition of S above).

If n is big (like $n = 100$), and we get a value that is smaller than -1.96 or greater than 1.96, that is "unusual" for the Standard Normal distribution, so we will not believe that the mean is indeed 172 cm. That means, we reject H_0. But if the value of the statistic calculated, falls within the range $[-1.96, 1.96]$, we can't reject the hypothesis H_0, as we do not have strong evidence against it.

Example: The breaking strains of reels of string produced at a certain factory have a standard deviation of 1.5 kg. A sample of 25 reels from a certain batch were tested and their mean breaking strain was 5.3 kg.

a) Find a 95% confidence interval for the mean breaking strain of string in this batch.

b) If the management states that the mean breaking strain of reels of string produced here is 5.5, would you accept it?

c) What size sample is required to obtain a 95% confidence interval of width at most 0.5?

Solution: a) Standard deviation of the mean is $\frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{25}}$. So the 95% confidence interval for the mean is $\bar{x} \pm 1.96 \times \frac{1.5}{\sqrt{25}} = 5.3 \pm 1.96 \times 0.3 = 5.3 \pm 0.588 = (4.712,\ 5.888)$.

b) 5.5 is within the above confidence interval. This means we don't have strong evidence against the mean breaking strain stated by the management, so we accept it.

c) We want the width of the confidence interval to be at most 0.5. That is, $2\times 1.96\times \frac{\sigma}{\sqrt{n}} = 2\times 1.96\times \frac{1.5}{\sqrt{n}} \leq 0.5$. Or equivalently, $2 \times 1.96 \times 1.5/0.5 \leq \sqrt{n}$. This implies $11.76^2 \leq n$, $139 \leq n$.

An example with two samples: An experiment was conducted to compare the drying properties of two paints, Quickdry and Speedicover. In the experiment, 200 similar pieces of metal were painted, 100 randomly allocated to Quickdry and the rest to Speedicover.

The table below summarises the times, in minutes, taken for these pieces of metal to become touch-dry.

	Quickdry	Speedicover
Mean	28.7	30.6
Standard deviation	7.32	3.51

It is believed that the time taken for paint to become touch-dry is Normally distributed.

The manufacturer of Quickdry claims that on average this paint takes 25 minutes to become touchdry.

1) Stating clearly your hypotheses and using a 1% significance level, test whether or not these data are consistent with the claim of the manufacturer.

2a) Using a 1% significance level, test whether or not the mean time for Quickdry to become touch-dry is *equal to* that for Speedicover.

2b) (*one-sided version*) Using a 1% significance level, test whether or not the mean time for Quickdry to become touch-dry is *less than* that for Speedicover. State your hypotheses clearly.

Solution: As the samples are large we can use the sample standard deviations instead of σ_1 and σ_2, the test statistic will have an approximately $N(0, 1^2)$ distribution.

1) $H_0: \quad \mu_1 = 25$. The alternative hypothesis is $H_1: \quad \mu_1 \neq 25$.

$n_1 = 100$ and $S_1 = 7.32$. The value of the test statistic is

$$z = \frac{28.7 - 25}{7.32/\sqrt{100}} = 5.05 \ .$$

Looking up in the Normal distribution table, the 1% (two-sided) critical values are ± 2.5758. This value of z is outside the acceptable region, so we reject H_0 and say that the data are not consistent with the manufacturer's claim.

2a) $H_0: \quad \mu_1 = \mu_2$. The alternative hypothesis is $H_1: \quad \mu_1 \neq \mu_2$.

$n_1 = 100$, $S_1 = 7.32$, $n_2 = 100$, $S_2 = 3.51$. The value of the test statistic (according to H_0, when $\mu_1 - \mu_2 = 0$) is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{28.7 - 30.6}{\sqrt{\frac{7.32^2}{100} + \frac{3.51^2}{100}}} = -2.341 \ .$$

As the 1% (two-sided) critical value is ± 2.5758 (because $\Phi(2.5758) = 0.995$), -2.341 is inside the acceptable region, so we accept H_0, the equality of the means.

2b) The right null-hypothesis is $H_0: \quad \mu_1 \geq \mu_2$; and the alternative hypothesis is $H_1: \quad \mu_1 < \mu_2$.

H_0 is equivalent to $\mu_1 - \mu_2$ being non-negative, so a Normally distributed random variable with mean $\mu_1 - \mu_2$ can not be smaller than a critical value. This is a one-sided test, the 1% critical value is -2.3263 (in the table we find $\Phi(+2.3263) = 0.99$). The value of the test statistic is the same as in 2a), we still use $\mu_1 - \mu_2 = 0$; but now with a one-sided confidence interval. -2.341 is smaller than the critical value, this is a strong evidence agains H_0. (We say the result is significant, and we can reject H_0 and conclude that there is evidence that the mean time for Quickdry to become touch-dry is less than the mean time for Speedicover.)

Note that the null hypothesis is the opposite of the statement proposed by the data. This is because if H_0 was $\mu_1 \leq \mu_2$, we could accept it without any calculations. (The statistic to be calculated would be the same, obviously yielding a negative number (as above). The 99% confidence interval would be $(-\infty, +2.3263]$, cutting off 1 percent from the positive side. A negative number falls within this interval without checking.) This would make the method meaningless.

Also note the difference between 2a) and 2b) $-$ the answer depends on whether our hypothesis is one- or two-sided! First we "proved" that $\mu_1 = \mu_2$, then we also "proved" that $\mu_1 < \mu_2$. In this case, if we used 5% level of significance, we would get consequent results: we would reject H_0 in both 2a) and 2b), getting $\mu_1 \neq \mu_2$ and $\mu_1 < \mu_2$ respectively.

Unfortunately you must get used to this phenomenon in statistics: your result may depend on what you choose for H_0. The terminology to describe the result when accepting H_0 is: "we don't have strong evidence against it". And even when we reject H_0, it is possible (with a 1-5% chance) that it is still true. Also, these methods are *inclined* to accept the null-hypothesis (only reject it if the data is strongly against it). So you put the safer statement to H_0, and the more risky statement to H_1. For example, H_0: the drug is useless, don't sell it to customers, H_1: the drug makes a real difference, people taking it feel so much better that it is worth spreading.

Other types of hypothesis tests: Goodness of fit and Independence

This is an other type of hypothesis testing, but the basic principle is the same: we calculate a numerical expression called *statistic*, and check whether it is acceptable according to the *distribution* the statistic should follow *when H_0 is true.*

In this problem type, we have nominal or discrete data. Results are collected in a table, where the cells correspond to categories, and the number of observations belonging to each category is written in the cells. First of all, if there are any cells with frequency less then 5, we must combine some cells so that the number of observations, the "observed" frequencies (O_i) reach 5 in each cell.

In these problems the *null-hypothesis* is, that the data has a given type of distribution (maybe having a parameter calculated from the data), which *implies "expected" frequencies* (E_i). If the discrepancies from these expected values are not big, we accept H_0. If the discrepancies are too big, we reject H_0. The statistic we calculate here is

$$\sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} ,$$

summing over all cells of the table (n =number of cells). Think about this expression as the sum of squared relative errors. When the null-hypothesis is true, this statistic has approximately *chi-squared* (χ^2) distribution. (The χ^2 distribution is defined by adding the squares of independent Standard Normal random variables. Adding k of them yields a random variable having χ^2 distribution with k degrees of freedom. The values of these distribution functions can also be looked up in tables, like those of the Standard Normal distribution.)

The squared relative errors we add here, are not completely independent. The number of degrees of fredom of the χ^2 distribution that this statistic follows is less than n. A major step in this topic is, to figure out the *number of degrees of freedom.* The general rule is, that the number of degrees of freedom is the *number off cells* the data is collected to as frequencies, *lowered by the number of constraints.* E.g. the total number of observations is fixed, this is always a constraint. Also, if one or more *parameters* of the fitted distribution (described in H_0) are to be calculated from the data, each calculated parameter gives rise to another constraint.

Summarising the above:

(i) We must combine cells in our data table so that the number of observations reach 5 in each cell;

(ii) if the distribution of the frequencies in the table is according to H_0, then $\sum_{i=1}^{n} \frac{(O_i-E_i)^2}{E_i} \sim \chi^2_v$ approximately, where v is the number of degrees of freedom;

(iii) $v =$ the number of cells after combining (n), minus the number of constraints.

E.g. if we are not fitting a distribution, but testing *independence* (which also determines expected frequencies) in a *contingency table* (crosstab), the row and column totals also define constraints. But the row totals and column totals add up to the same number, the big total, so they are not independent constraints. So the number of degrees of freedom for an $r \times c$ table is $v = rc - r - (c-1) = rc - r - c + 1 = (r-1)(c-1)$.

Example for fitting a distribution:

Suppose you take a die and roll it $N = 120$ times. If the die is *unbiased* you would in theory expect each of the numbers 1 to 6 to appear 20 times. In other words, you would expect the frequencies of the numbers to fit the discrete uniform distribution:

k	1	2	3	4	5	6
$P(X = k)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

For 120 rolls the expected frequencies would each be $P(X = k) \times 120 = \frac{1}{6} \times 120 = 20$. But in a sample you are taking, it would be surprising if the observed frequency for each number was exactly 20. Suppose now that the die was *biased*, this time you would not expect the observed frequencies of the numbers to be equal even in the long run.

You may get results like these:

Number, k	1	2	3	4	5	6
Observed frequency of k	23	15	25	18	21	18

How can you tell if the die is biased? Could these results come from an unbiased die?

Solution:

H_0 : unbiased die, discrete uniform distribution over $\{1, 2, 3, 4, 5, 6\}$.

The statistic to be calculated is $\sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} = \frac{9}{20} + \frac{25}{20} + \frac{25}{20} + \frac{4}{20} + \frac{1}{20} + \frac{4}{20} = \frac{68}{20} = 3.4$. Having 6 cells, and a known distribution (with no parameter estimated from the data) we compare this value with the critical values of the chi-squared distribution with $6 - 1 = 5$ degrees of freedom. Since

$$F_{\chi^2_5}(11.070) = 0.95 \ ,$$

the relevant critical value (yielding a 95% confidence interval) is 11.070, and we accept the value of the statistic if it falls between 0 and 11.070. (We have a one-sided problem here since we only care about whether the sum of squared relative errors is too big.) Since $3.4 < 11.070$, we accept the null-hypothesis (unbiased die) on a 5% significance level: there is not enough evidence to suggest that the die is biased. (Still, if the die is biased, you can get enough evindence with a bigger number of trials.)

Here we used a distribution without free parameters. If you have to use a more difficult distribution, like Binomial or Poisson, without its parameter(s) given, first you must find the parameter(s). For Poisson, its parameter (λ) is equal to the mean, so your estimate for λ will be the average. For the Binomial, n can be seen from tha data, and p is calculated from the average again: the mean is np, so your estimate for p will be the average divided by n.

If there are cells with frequency below 5, join the low-frequency cells in the whole calculation, so that each (combined) cell has frequency at least 5. (Don't forget about unfilled categories, meaning cells not shown: e.g. the possible values of Poisson distribution are not bounded, but surely the outcomes in an experiment do have a maximum. So unless you create an unbounded category at the top, you will have hidden empty categories!)

Example for testing independence in a contingency table: You study the frequency with which A-level Maths passes at grades A, B, C occur. You may also be interested in which of the two schools the students attended. Here you have two criteria: the pass level and the school. You can show these results by means of a *contingency table*, which shows the frequency with which each of the results occurred at each school separately.

	A	B	C	Totals
School 1	18	12	20	50
School 2	26	12	32	70
Totals	44	24	52	120

This is called a 2×3 contingency table. The question that arises is, "Is there any difference between the two schools' sets of results?" You pose the hypothesis "Are the two criteria independent?"

Solution:

H_0: school and grade of pass are independent.

H_1: school and grade of pass are not independent.

For example, $P(\text{grade A}) = \frac{44}{120}$, $P(\text{School 1}) = \frac{50}{120}$, so assuming H_0 is true, $P(\text{grade A and School 1}) = P(\text{grade A}) \times P(\text{School 1}) = \frac{44}{120} \times \frac{50}{120}$. The expected frequency of passes at A from *School* 1 is therefore $\frac{44}{120} \times \frac{50}{120} \times 120 = 18.33$.

In general, expected frequency = row total × column total / grand total.

Now we can calculate the statistic $\sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i}$ which must have the χ^2 distribution with degree of freedom $(2-1) \times (3-1) = 2$, in case of independence. The value of the statistic is 0.9165. From tables, the critical value for the 5% significance level is 5.991, and in the interval $(0,\ 5.991)$ we accept the result to come from χ^2_2 distribution. This is the case now, so we accept H_0 : school and grade of pass are independent.